Navyug Vidyabhavan Trust

C. K. Pithawala College of Engineering & Technology

Information Technology Engineering Department
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(Project Based Learning)

Computer Network (BE04000271) BE II, Sem IV

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MULTI-ROUTER CAMPUS NETWORK USING OSPF (SINGLE AREA)

INTRODUCTION

- A multi-router campus network is a network design where multiple routers are used to connect different departments within a campus.
- Each department (like Library, Lab, Admin, etc.) has its own local network (LAN) connected through switches.
- Routers are used to interconnect these LANs, enabling communication between different departments.
- In large networks, manual routing becomes difficult, so dynamic routing protocols are used.
- OSPF is a dynamic routing protocol that automatically finds the shortest path for data transmission.
- In a single-area OSPF network (Area 0), all routers belong to one area, which simplifies configuration and is suitable for small to medium campus networks.
- OSPF uses a link-state algorithm to share network information and build accurate routing tables.

PROBLEM STATEMENT

- In a campus environment, multiple departments need to communicate efficiently with each other for smooth data sharing and operations. However, without a proper network design, communication can become slow and unreliable, making it difficult to share information across departments. Additionally, the lack of routing mechanisms may lead to network isolation, where different departments cannot interact effectively. To overcome these issues, there is a need for proper IP addressing, subnetting, and dynamic routing. Therefore, a campus network is designed using the OSPF to ensure efficient, reliable, and optimized communication.

OBJECTIVES

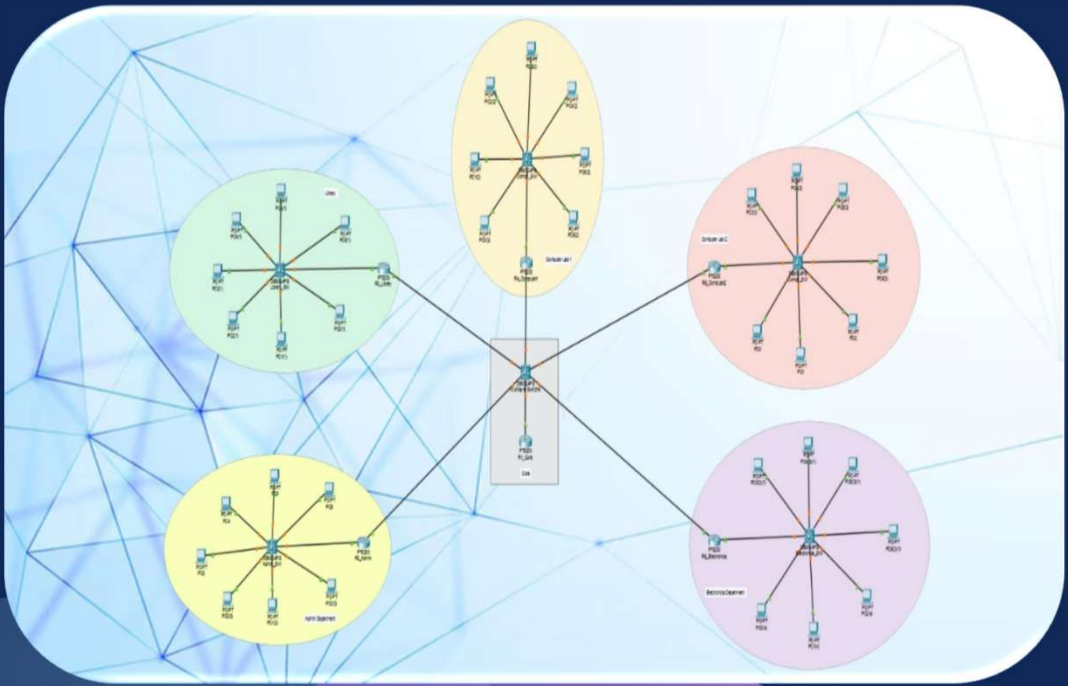
- To design a structured and scalable campus network infrastructure.
- To implement dynamic routing using OSPF.
- To apply efficient IP addressing and subnetting techniques.
- To ensure and verify communication between different departments through simulation.
- To test and evaluate network performance using standard commands such as Ping and Tracert.

SCENARIO DESCRIPTION

- The network represents a campus environment connecting multiple departments such as Library, Computer Labs, Admin, and Electronics.
- Each department has several PCs connected through a switch, forming a local area network (LAN).
- These departmental networks are connected using routers, which enable communication between different departments.
- All routers are further connected to a central core device that acts as the backbone of the network.
- OSPF is used to dynamically exchange routing information and select the shortest path for data transmission.
- This design ensures efficient communication, scalability, and reliable performance of the campus network.

NETWORK TOPOLOGY

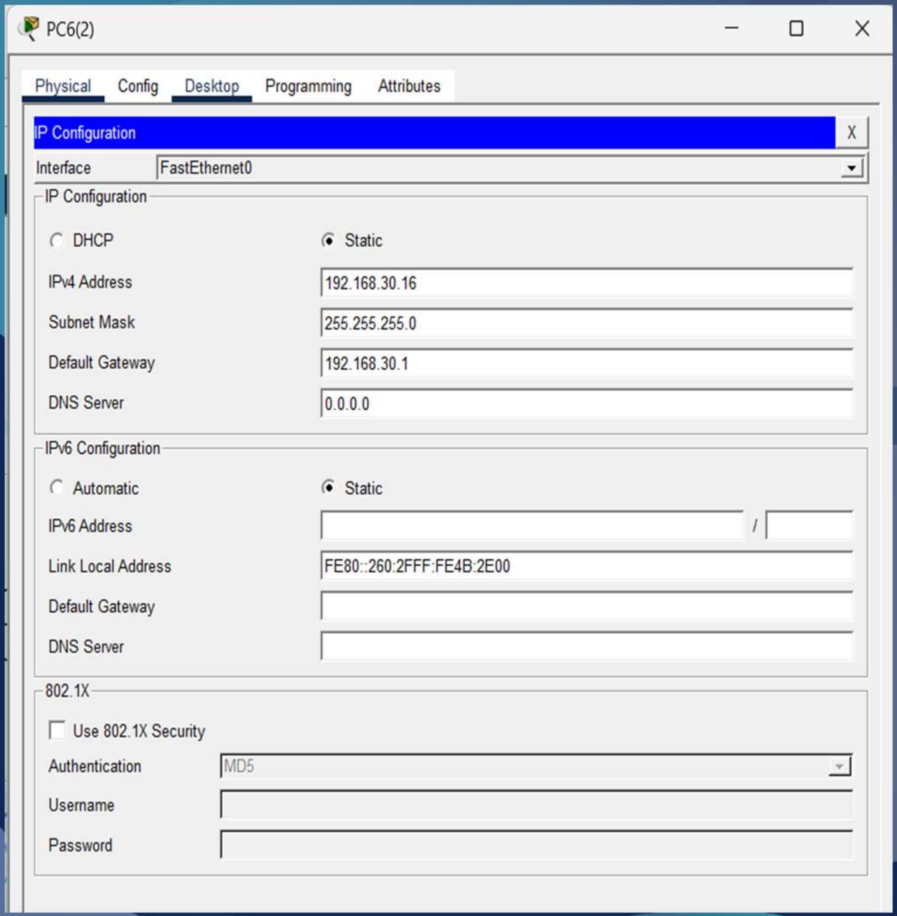
- The network uses a **Hybrid Topology**, which is a combination of different topologies.
- **Star Topology (Within Departments):**
  - In each department (Library, Labs, Admin, Electronics), all PCs are connected to a central switch.
  - This forms a **star topology**, where the switch acts as the central device.
- **Tree Topology (Overall Network):**
  - All departmental routers are connected to a **central core device.**
  - This creates a **tree (hierarchical) structure** with multiple layers.

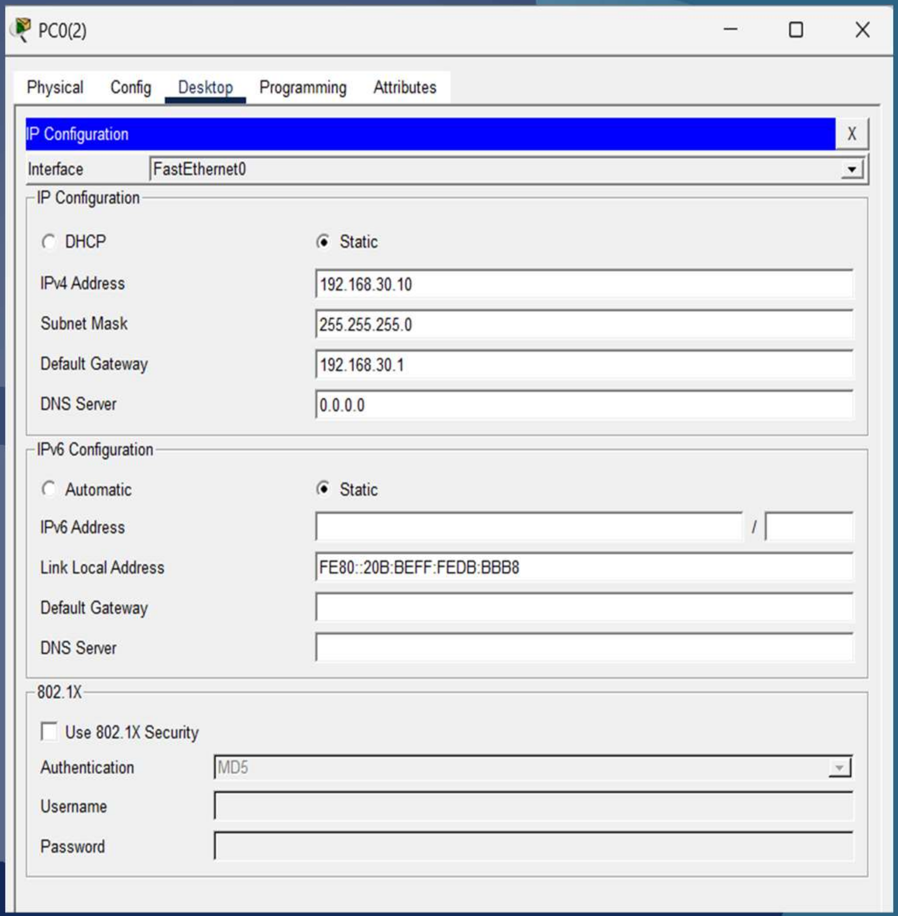


IP ADDRESSING TABLE

Department	Network Address	Subnet Mask	Default Gateway
Admin	192.168.10.0	255.255.255.0	192.168.10.1
Library	192.168.20.0	255.255.255.0	192.168.20.1
Comp Lab 1	192.168.30.0	255.255.255.0	192.168.30.1
Comp Lab 2	192.168.40.0	255.255.255.0	192.168.40.1
Electronics	192.168.50.0	255.255.255.0	192.168.50.1

IP ADDRESSING





TECHNOLOGIES USED

- **OSPF Routing:**
  Used for dynamic routing between routers, automatically sharing routes and selecting the shortest path (Area 0) for efficient communication.

- **Switching (LAN):**
  Switches connect multiple PCs within each department and enable communication using MAC addresses.

- **IP Addressing & Subnetting:**
  Each department has a separate subnet (192.168.x.x), improving network management, performance, and reducing congestion.

- **Hierarchical Design:**
  Follows a 3-layer architecture:
  Access (PCs & switches), Distribution (routers), Core (central device) for scalability and organization.

CONFIGURATION HIGHLIGHTS

- **Router Setup**
  - enable
  - configure terminal
  - router ospf 1
  - network 192.168.10.0 0.0.0.255 area 0
- **Verification**
  - show ip route
  - show ip protocols
- **Interface Setup**
  - interface GigabitEthernet0/0/0
  - ip address 192.168.10.1 255.255.255.0
  - no shutdown

CONFIGURATION HIGHLIGHTS

```
R4_CompLab1>en
R4_CompLab1#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R4_CompLab1(config)#router ospf 1
R4_CompLab1(config-router)#network 192.168.10.0 0.0.0.255 area 0
R4_CompLab1(config-router)#
```

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Top

```
R5_CompLab2>en
R5_CompLab2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/24 is directly connected, GigabitEthernet0/0/0
L       10.0.0.14/32 is directly connected, GigabitEthernet0/0/0
O    192.168.10.0/24 [110/2] via 10.0.0.11, 4294967275:4294967244:4294967265,
GigabitEthernet0/0/0
O    192.168.20.0/24 [110/2] via 10.0.0.12, 4294967275:4294967244:4294967265,
GigabitEthernet0/0/0
O    192.168.30.0/24 [110/2] via 10.0.0.13, 4294967275:4294967244:4294967265,
GigabitEthernet0/0/0
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0/1
O    192.168.50.0/24 [110/2] via 10.0.0.15, 4294967275:4294967244:4294967265,
GigabitEthernet0/0/0

R5_CompLab2#
```

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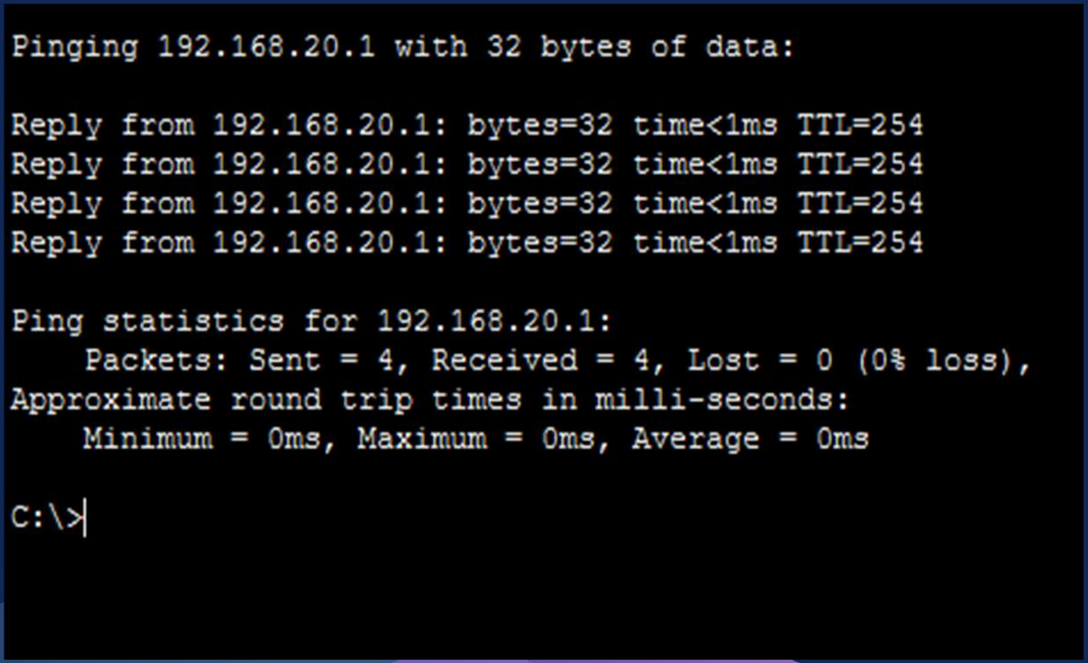
TESTING & RESULTS

- Ping Test
  Successful communication between networks
  All routers and PCs reachable

- Results
  No packet loss observed
  Network connectivity verified

- Output
  Ping command screenshots attached
  Routing table confirms OSPF routes



```
Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=254
Reply from 192.168.20.1: bytes=32 time<1ms TTL=254
Reply from 192.168.20.1: bytes=32 time<1ms TTL=254
Reply from 192.168.20.1: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

CONCLUSION

- The project demonstrates the design and implementation of a campus network using the OSPF routing protocol, connecting different departments through routers and switches.
- OSPF enables dynamic routing, allowing routers to exchange information automatically and select the most efficient path. The hierarchical topology improves scalability and network management.
- Using Cisco Packet Tracer, the project provides practical knowledge of IP addressing, routing, and network design. Testing confirms that the network works efficiently and reliably.
- Overall, the project reflects real-world networking concepts and helps in understanding modern network technologies.

THANK YOU